

Lecture 7, Jan 06, 2025

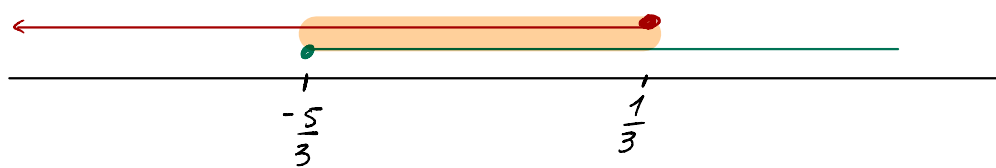
Problem 1. Solve the following inequalities. Display your answer in interval notation.

a). $|2 + 3x| \leq 3$

Sln. $-3 \leq 2 + 3x$ and $2 + 3x \leq 3$.

$$-5 \leq 3x \quad \text{and} \quad 3x \leq 1$$

$$-\frac{5}{3} \leq x \quad \text{and} \quad x \leq \frac{1}{3}$$



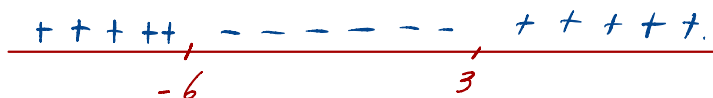
$$S = [-\frac{5}{3}, \frac{1}{3}]$$

b). $x^2 + 3x - 10 \geq 8$.

Sln. $x^2 + 3x - 10 - 8 \geq 0$

$$x^2 + 3x - 18 \geq 0$$

$$(x + 6)(x - 3) \geq 0$$



$$S = (-\infty, -6] \cup [3, \infty)$$

Problem 2. Find the limit if it exists. If the limit doesn't exist explain why.

a). $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x^2 + x - 6}$

Sln. $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x^2 + x - 6} = \lim_{x \rightarrow 2} \frac{(x-2)(x+2)}{(x-2)(x+3)} = \lim_{x \rightarrow 2} \frac{x+2}{x+3} = \frac{4}{5}$

$$b). \lim_{x \rightarrow \frac{1}{2}} \frac{\sqrt{2x+3} - 2}{2x-1}$$

Sln.

$$\begin{aligned} \lim_{x \rightarrow \frac{1}{2}} \frac{\sqrt{2x+3} - 2}{2x-1} &\times \frac{\sqrt{2x+3} + 2}{\sqrt{2x+3} + 2} = \lim_{x \rightarrow \frac{1}{2}} \frac{2x+3-4}{(2x-1)(\sqrt{2x+3}+2)} \\ &= \lim_{x \rightarrow \frac{1}{2}} \frac{\cancel{2x-1}}{(\cancel{2x-1})(\sqrt{2x+3}+2)} \\ &= \lim_{x \rightarrow \frac{1}{2}} \frac{1}{\sqrt{2x+3}+2} = \frac{1}{4} \end{aligned}$$

$$c). \lim_{x \rightarrow 4^-} \frac{|x-4|}{3x^2-3x-36}$$

Sln.

$$\begin{array}{c} | \\ \hline x \quad 4 \\ | \\ x-4 < 0 \end{array}$$

So, $|x-4| = -(x-4)$. by definition of abs. value.
 $= -x+4$

$$\begin{aligned} \lim_{x \rightarrow 4^-} \frac{|x-4|}{3(x^2-x-12)} &= \lim_{x \rightarrow 4^-} \frac{-\cancel{(x-4)}}{3(\cancel{x-4})(x+3)} \\ &= \lim_{x \rightarrow 4^-} \frac{-1}{3(x+3)} = -\frac{1}{21} \end{aligned}$$

Problem 3. Suppose that $f(x) = \frac{3}{x}$

a). Show that $f'(x) = -\frac{3}{x^2}$ using the limit definition of derivative.

Sln.

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\frac{3}{x+h} - \frac{3}{x}}{h} \\ &= \lim_{h \rightarrow 0} \frac{\frac{3x - 3(x+h)}{x(x+h)}}{\frac{h}{1}} \end{aligned}$$

$$= \lim_{h \rightarrow 0} \frac{3x - 3x - 3h}{xh(x+h)}$$

$$= \lim_{h \rightarrow 0} \frac{-3\cancel{h}}{x\cancel{h}(x+h)}$$

$$= \lim_{h \rightarrow 0} \frac{-3}{x(x+h)} = \frac{-3}{x(x+0)} = \frac{-3}{x^2}$$

b). Find the equation of the line tangent to f at the point $x=6$.

Sln. Point: $(6, f(6))$ $f(6) = \frac{3}{6} = \frac{1}{2} \Rightarrow (6, \frac{1}{2})$.
 x_1 y_1

slope: $m = f'(6) = \frac{-3}{(6)^2} = \frac{-3}{36} = -\frac{1}{12}$.

$$y - y_1 = m(x - x_1)$$

$$y - \frac{1}{2} = -\frac{1}{12}(x - 6)$$

$$y = -\frac{1}{12}x + 1.$$

Problem 4. Compute

$$\frac{d}{dx} [(2x^2 - 3)(5 - 2x)].$$

Sln. Notice that $\frac{d}{dx} [f(x) \cdot g(x)]$ where $f(x) = 2x^2 - 3$ $g(x) = 5 - 2x$.

we can apply the product rule:

$$\frac{d}{dx} [f(x) \cdot g(x)] = f'(x) \cdot g(x) + f(x) \cdot g'(x).$$

$$f'(x) = \frac{d}{dx} (2x^2 - 3) = \frac{d}{dx} (2x^2) - \frac{d}{dx} (3) = 2 \frac{d}{dx} (x^2) - \frac{d}{dx} (3)$$

$$= 2(2x) - 0 = 4x.$$

$$g'(x) = -2.$$

$$\begin{aligned}\frac{d}{dx} [(2x^2-3)(5-2x)] &= 4x(5-2x) + (2x^2-3)(-2) \\ &= 20x - 8x^2 - 4x^2 + 6 \\ &= -12x^2 + 20x + 6.\end{aligned}$$

Problem 5 Let

$$f(x) = \begin{cases} x^2+1 & \text{if } x \leq 0 \\ \frac{\sin x}{x} & \text{if } 0 < x < 2 \\ 0 & \text{if } 2 \leq x. \end{cases}$$

a). Find, with explanation, all real numbers a , s.t. $f(x)$ is not continuous at a .

Sln. Problems when $x=0$, $x=2$.

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (x^2+1) = 1.$$

$\Rightarrow f$ is continuous at $x=0$.

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{\sin x}{x} = 1$$

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} \frac{\sin x}{x} = \frac{\sin 2}{2}.$$

$$\text{Since } \lim_{x \rightarrow 2^-} f(x) \neq \lim_{x \rightarrow 2^+} f(x)$$

f is not continuous at $x=2$.

$$\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} 0 = 0$$

Problem 6. $\lim_{x \rightarrow -3^-} \frac{3x+1}{x^2+3x}$

$$\frac{\quad}{-3}$$

Sln.

$$\text{So, } \lim_{x \rightarrow -3^-} \frac{3x+1}{x^2+3x} = -\infty.$$

$$\frac{3(-4)+1}{(-4)^2-12} = \frac{-4}{16-12} = -\frac{11}{4}.$$